

KILLARA HIGH SCHOOL



YEAR 12 – 2008

HSC ASSESSMENT 2

MATHEMATICS EXTENSION 1

Time allowed: 2 hours

Total marks: 84

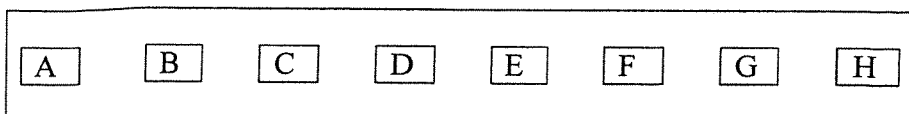
- **NOT ALL** questions are of equal value.
- Start each question in a new booklet.
- Show all necessary working.

Question 1 (Begin a new booklet)**Marks**

- a) $(x - 2)$ is a factor of the polynomial $P(x) = 2x^3 + x + a$. Find the value of a . 1
- b) $\int_{-1}^1 (1 + 5x)^4 dx$ 2
- c) On a number plane, indicate the region defined by $y \geq |2x| + 1$ and $y < 3$. 3
- d) Find the acute angle (to the nearest degree) between the line $y - x + 1 = 0$ and the curve $y = (x - 1)^2$ at the point of intersection where $x = 2$. 3
- e) Solve the inequality $\frac{2x+1}{x-1} \geq 3$. 3

Question 2 (Begin a new booklet)**Marks**

a)



A security lock has 8 buttons labeled as shown. Each person using the lock is given a 3-letter code.

- i. How many different codes are possible if letters can be repeated and their order is important? 1
- ii. How many different codes are possible if letters can not be repeated and their order is important? 1
- iii. Suppose that the lock operates by holding 3 buttons down together, given that the order is NOT important. How many different codes are possible? 1

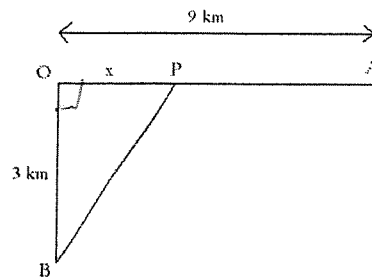
Question 2 (continue)

- a) Consider the equation $x^3 + 6x^2 - x - 30 = 0$.

One of the roots of this equation is equal to the sum of the other two roots. Find the values of the three roots.

4

b)



The diagram shows a straight road OA, running due east. A motorist is in open country at B, 3 km due south of O. He must reach the point A as quickly as he can, where $OA = 9$ km. The car can travel at 80 km/h in open country and at 100 km/h along the road. He intends to proceed in a straight line to some point P on the road and then travel along the road.

- i. Let x be the distance OP, show that the time taken to reach the point A is given by

$$T = \frac{\sqrt{x^2 + 9}}{80} + \frac{9 - x}{100} \quad \mathbf{2}$$

- ii. Show that the minimum time for the entire journey is $6 \frac{3}{4}$ min.

4

Question 3 (Begin a new booklet)

Marks

- a) Prove the identity $\tan (45^\circ + A) + \tan (45^\circ - A) = \frac{2}{\cos 2A}$ **3**
- b) The function $f(x)$ is defined by $f(x) = \frac{1}{x^2 + 1}$.
- i. Show that $f(x)$ is an even function. **1**
 - ii. State the domain of $f(x)$. Find the intercepts of the curves $f(x)$ with the co-ordinate axes. **2**
 - iii. Write down the horizontal asymptotes. **1**
 - iv. Find and determine the nature of the turning point. **3**
 - v. Describe the behavior of $f(x)$ as $x \rightarrow \infty$ and $x \rightarrow -\infty$ **1**
 - vi. Draw a sketch of $y = f(x)$. **2**

Question 4 (Begin a new booklet)**Marks**

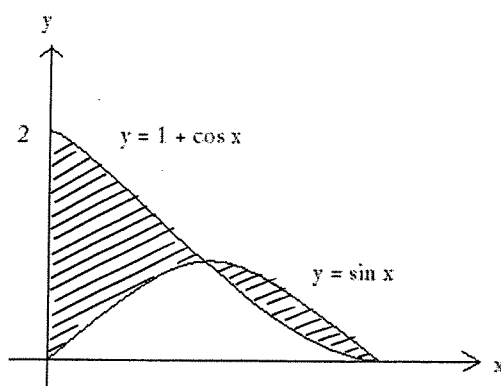
a) Find $\int \frac{x}{\sqrt{1+x}} dx$. Using substitution $u = 1+x$.

4

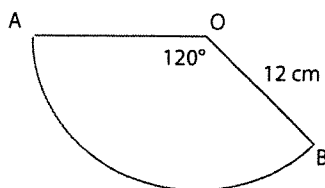
b) i. By **solving** for x , show that the solutions to the equation $\cos x - \sin x = -1$ for $0 \leq x \leq \pi$ are $x = \frac{\pi}{2}$ and $x = \pi$.

3

ii. Hence find the shaded area bounded by the graphs of $y = \sin x$ and $y = 1 + \cos x$ (shown below), from $x = 0$ and $x = \pi$.

3

c)



A piece of paper is cut in the shape of a sector of a circle. The radius is 12 cm and the angle at the centre is 120° . The straight edges of the sector are placed together so that a cone is formed.

- Show that the base of the cone has radius 4 cm.
- Show that the cone has perpendicular height $8\sqrt{2}$ cm.
- Hence find, the exact form, the volume of the cone.
- Find the curved surface area of the cone.

1**1****1****1**

Question 5 (Begin a new booklet)**Marks**

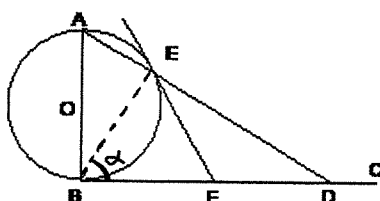
- a) Find $\lim_{x \rightarrow 0} \frac{\sin 6x}{5x}$. **2**
- b) Find $\frac{dy}{dx}$ for the following:
- i. $y = \tan \frac{x}{2} + \sin^3 x$ **2**
- ii. $y = e^{-x} \sin x$ **2**
- c) Find:
- i. $\int \cos\left(2x - \frac{\pi}{3}\right) dx$ **1**
- ii. $\int \sin^2\left(\frac{x}{2}\right) dx$ **2**
- d) On the same axes, sketch the curves $y = \sin 2x$, $y = \cos x$ and $y = \cos x + \sin 2x$, for $0 \leq x \leq \pi$. **4**

Question 6 (Begin a new booklet)**Marks**

- a) Given that $\log_2 5 = 2.32$ and $\log_2 10 = 3.32$, find $\log_2 0.025$. 2
- b) Differentiate with respect to x
- i. $\ln(3x^2 - 1)$ 1
- ii. $\frac{2x+1}{e^x}$ 2
- c) Find $\int \frac{2x-1}{x+3} dx$ 2
- d) Find the exact volume of the solid formed when the region bounded by the x -axis and the curve $y = e^x + e^{-x}$, between $x = -1$ and $x = 1$, is rotated around the x -axis. 4

Question 7 (Begin a new booklet)**Marks**

- a) Find the coordinates of the focus and the vertex of the parabola $x^2 = -12y + 24$. 2



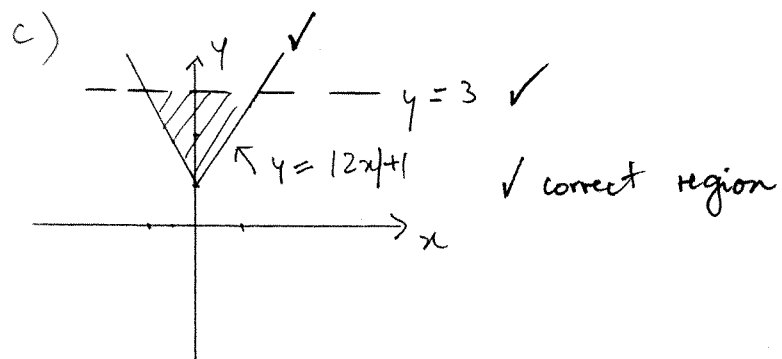
- b) In the diagram, AB is a diameter of the circle, centre O , and BC is tangential to the circle at B . The line AED intersect the circle at E and BC at D . The tangent to the circle at E intersects BC at F . Let $\angle EBF = \alpha$ radian.
- i. Copy the diagram into your writing booklet.
- ii. Prove that $\angle FED = \frac{\pi}{2} - \alpha$. 4
- iii. Prove that $BF = FD$. 2

Q1

(i) a) $P(2) = 2 \times 2^3 + 2 + a = 0$

$a = -18$ ✓

b) $\int_{-1}^1 (1+5x)^4 dx = \left[\frac{1}{5} [1+5x]^5 \right]_{-1}^1$ ✓
 $= \frac{1}{5} (6^5 - (-4)^5) = 352$ ✓



d) $y_1 = x - 1, m_1 = 1$

$y_2 = x^2 - 2x + 1$

$y'_2 = 2x - 2$ at $x = 2$

$m_2 = 2$ ✓

$\tan \theta = \left| \frac{1-2}{1+1 \times 2} \right| = \frac{1}{3}$ ✓

$\theta \doteq \circ$ ✓

e) $(2x+1)(x-1) \geq 3(x-1)$ ✓ $x \neq 1$

$3(x-1)^2 - (2x+1)(x-1) \leq 0$

$(x-1)(3x-3-2x-1) \leq 0$

$(x-1)(x-4) \leq 0$ ✓

$1 < x \leq 4$ is the solution ✓

12 marks

No half mark

Q2

(i) a) $8^3 = 512$ ✓

b) $8 \times 7 \times 6 = 8P_3$
 $= 336$ ✓

c) $8C_3 = 56$ ✓

(ii)

let the roots be α, β, γ

$\alpha = \beta + \gamma$

$2\beta + 2\gamma = -6$ ✓

$\beta + \gamma = -3 \Rightarrow \beta = -3 - \gamma$ (1)

$\beta \cdot \gamma \cdot (\beta + \gamma) = +30$

$-3\beta\gamma = +30$ ✓

$\beta\gamma = -10 \leftarrow \text{Sub (1) into}$

$\Rightarrow (-3 - \gamma)\gamma = -10$

$\gamma^2 + 3\gamma - 10 = 0$

$(\gamma + 5)(\gamma - 2) = 0$ ✓

$\gamma = -5, 2$

$\alpha = -3$

$\therefore$ The 3 roots are : 2, -3 & -5 ✓

(iii)

a) $BP = \sqrt{x^2 + 9}$ $PA = 9 - x$ ✓

The time taken to travel from :

$B \rightarrow P$ is $\frac{\sqrt{x^2 + 9}}{80}$ and

$P \rightarrow A$ is $\frac{9 - x}{100}$ ✓

$\therefore$ Total time for the Journey :

$T = \frac{\sqrt{x^2 + 9}}{80} + \frac{9 - x}{100}$

b) $\frac{dT}{dx} = \frac{2x}{2 \times 80 \sqrt{x^2 + 9}} - \frac{1}{100}$

$= \frac{x}{80 \sqrt{x^2 + 9}} - \frac{1}{100} = 0$ ✓

$\Rightarrow 10x - 8\sqrt{x^2 + 9} = 0$

$100x^2 = 64x^2 + 576$

$36x^2 = 576$

$x^2 = 16$ ✓

$x = 4$ ✓

$\therefore x = 4$ is a min.

when $x = 4$

$T = \frac{\sqrt{16 + 9}}{80} + \frac{9 - 4}{100} = 0.1125 \text{ h}$

$= 6\frac{3}{4} \text{ min.}$ ✓

13 marks

Q3

$$(i) \tan(45^\circ + A) + \tan(45^\circ - A) = \frac{2}{\cos 2A}$$

$$\text{L.H.S} = \frac{1 + \tan A}{1 - \tan A} + \frac{1 - \tan A}{1 + \tan A}$$

$$= \frac{(1 + \tan A)^2 + (1 - \tan A)^2}{1 - \tan^2 A}$$

$$= \frac{(\cos A + \sin A)^2}{\cos^2 A} + \frac{(\cos A - \sin A)^2}{\cos^2 A}$$

$$\frac{\cos^2 A - \sin^2 A}{\cos^2 A}$$

$$= \frac{\cos^2 A + 2 \cancel{\cos A \sin A} + \sin^2 A + \cos^2 A - 2 \cancel{\sin A \cos A} + \sin^2 A}{\cos^2 A}$$

$$= \frac{2(\cos^2 A + \sin^2 A)}{\cos 2A} = \frac{\cos 2A}{\cos 2A} = \text{R.H.S}$$

$$(ii) a) f(-x) = \frac{1}{(-x)^2 + 1} = \frac{1}{x^2 + 1} = f(x)$$

$$b) \text{ when } x = 0 \quad f(x) = \frac{1}{0 + 1} = 1$$

domain: all real x .

$$c) \lim_{x \rightarrow \infty} \frac{1}{x^2 + 1} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x^2}}{\frac{1}{x^2} + 1} = 1$$

$y = 1$ is horizontal asymptote.

$$d) f(x) = -2x(x+1)$$

$$= \frac{-2x}{(x^2 + 1)^2} = 0$$

$$\Rightarrow x = 0, y = 1$$

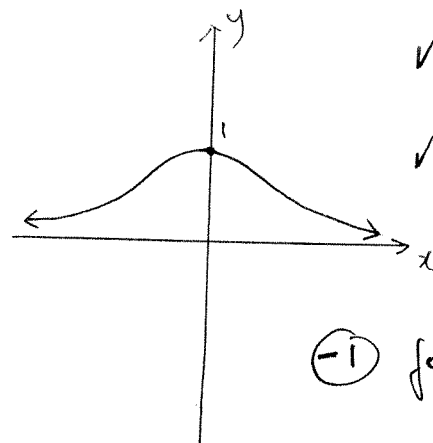
x	-1	0	1
$f'(x)$	$\nearrow$	0	$\searrow$

$\therefore (0, 1)$ is a min. point.

$$e) \text{ as } x \rightarrow \infty, \quad x \rightarrow -\infty$$

$$f(x) \rightarrow 0, \quad f(x) \rightarrow 0$$

f)



✓ correct shape

✓ labelling

⊖ for any missing

12 marks

Q4

$$(i) \int \frac{x}{\sqrt{1+x}} dx$$

$$u = 1+x \Rightarrow x = u-1$$

$$du = dx$$

$$\int \frac{x}{\sqrt{1+x}} dx = \int \frac{u-1}{\sqrt{u}} du \quad \checkmark$$

$$= \int (u-1) u^{-\frac{1}{2}} du$$

$$= \int u^{\frac{1}{2}} - u^{-\frac{1}{2}} du \quad \checkmark$$

$$= \frac{2}{3} u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \quad \checkmark$$

$$= \frac{2}{3} (x+1) \sqrt{x+1} - 2\sqrt{x+1} + C \quad \checkmark$$

$$(ii) \cos x - \sin x = -1$$

$$r = \sqrt{1+1} = \sqrt{2} \quad \checkmark$$

$$\tan \theta = \frac{1}{1} \Rightarrow \theta = \frac{\pi}{4}$$

$$\therefore \sqrt{2} \cos\left(x + \frac{\pi}{4}\right) = -1 \quad \checkmark$$

$$\cos\left(x + \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

$$x + \frac{\pi}{4} = \frac{3\pi}{4}, \frac{5\pi}{4} \quad \checkmark$$

$$\begin{aligned} b) A &= \int_0^{\frac{\pi}{2}} 1 + \cos x - \sin x + \int_{\frac{\pi}{2}}^{\pi} \sin x - 1 - \cos x dx \quad \checkmark \\ &= \left[x + \sin x + \cos x \right]_0^{\frac{\pi}{2}} + \left[-\cos x - x - \sin x \right]_{\frac{\pi}{2}}^{\pi} \quad \checkmark \\ &= \frac{\pi}{2} + 1 - 1 + 1 - \pi - \left(-\frac{\pi}{2} - 1\right) = 2 \quad \checkmark \end{aligned}$$

(iii)

$$a) l_{AB} = 12 \times \frac{120 \times \pi}{180} = 8\pi$$

$$8\pi = 2\pi r \Rightarrow r = \frac{8\pi}{2\pi} = 4 \text{ cm} \quad \checkmark$$

b)

$$\begin{aligned} h &= \sqrt{12^2 - 4^2} \\ &= \sqrt{128} \\ &= 8\sqrt{2} \text{ cm} \quad \checkmark \end{aligned}$$

c)

$$\begin{aligned} V &= \frac{1}{3} \pi \times 4^2 \times 8\sqrt{2} \\ &= \frac{128\pi \sqrt{2}}{3} \pi \text{ cm}^3 \quad \checkmark \end{aligned}$$

d)

$$A_{\text{sector}} = \frac{1}{2} \times 12^2 \times \frac{120 \times \pi}{180} = 48\pi \text{ cm}^2 \quad \checkmark$$

14 marks

Q5. 15 marks

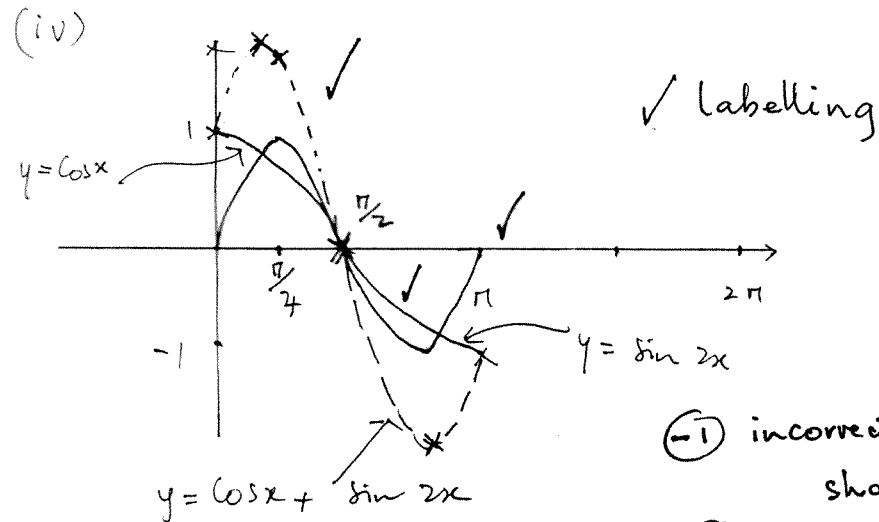
(i) $\lim_{x \rightarrow 0} \frac{\sin 6x}{5x} = \lim_{x \rightarrow 0} \frac{6}{5} \frac{\sin 6x}{6x} \checkmark$
 $= \frac{6}{5} \checkmark$

(ii) a) $y' = \frac{1}{2} \sec^2 \frac{x}{2} + 3 \sin^2 x \cos x \checkmark$

b) $y' = -e^{-x} \sin x + e^{-x} \cos x \checkmark$

(ii) a) $\int \cos \left(2x - \frac{\pi}{3} \right) dx = \frac{1}{2} \sin \left(2x - \frac{\pi}{3} \right) + C \checkmark$

b) $\int \sin^2 \frac{x}{2} dx = \int \frac{1}{2} - \frac{1}{2} \cos x dx \checkmark$
 $= \frac{x}{2} - \frac{1}{2} \sin x + C \checkmark$



⊖ incorrect scale/
shape

Q6. 11 marks

(i) $\log_2 0.025 = \log_2 \frac{25}{1000} = 2 \log_2 5 - 3 \log_2 10 \checkmark$
 $= 2 \times 2.32 - 3 \times 3.32 \checkmark$
 $= -5.32$

(ii) a) $\frac{d}{dx} \ln(3x^2 - 1) = \frac{6x}{3x^2 - 1} \checkmark$

b) $\frac{2e^x - e^x(2x+1)}{e^{2x}} \checkmark$
 $= \frac{e^x(2 - 2x - 1)}{e^{2x}} = \frac{1 - 2x}{e^x} \checkmark$

(iii) $\int \frac{2x-1}{x+3} dx = \int \frac{2(x+3)}{x+3} - \frac{7}{x+3} dx \checkmark$
 $= 2x - 7 \ln(x+3) + C \checkmark$

(iv) $y^2 = (e^x + e^{-x})^2 = e^{2x} + 2 + e^{-2x} \checkmark$

$V = \pi \int_{-1}^1 e^{2x} + 2 + e^{-2x} dx$

$= \pi \left[\frac{e^{2x}}{2} + 2x - \frac{e^{-2x}}{2} \right]_{-1}^1 \checkmark$

$= \pi \left(\frac{e^2}{2} + 2 - \frac{e^{-2}}{2} - \frac{e^{-2}}{2} + 2 + \frac{e^2}{2} \right) \checkmark$
 $= \pi (e^2 - e^{-2} + 4)$

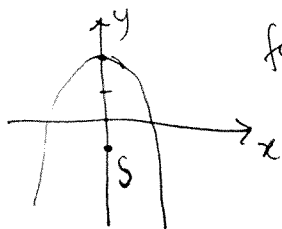
$$(i) \cdot x^2 = -12(y-2)$$

$$= -4 \times 3(y-2)$$

$$\text{Vertex: } (0, 2) \quad \checkmark$$

$$\text{focus: } (0, -1) \quad \checkmark$$

8 marks



(ii)

b) $BF = FE$ (tangents to the circle from a point) $\checkmark$

$\therefore \triangle BEF$ is isosceles. $\checkmark$

hence $\angle BEF = \alpha$

$\angle AEB = 90^\circ$ (angle in a semi circle) $\checkmark$

$$\therefore \angle BED = \pi - \frac{\pi}{2} = \frac{\pi}{2}$$

$$\begin{aligned} \angle FED &= \angle BED - \angle BEF \\ &= \frac{\pi}{2} - \alpha \end{aligned} \quad \checkmark$$

c)

$$\begin{aligned} \angle EDB &= \pi - \left(\frac{\pi}{2} + \alpha \right) \text{ (angle sum of } \triangle \text{)} \\ &= \frac{\pi}{2} - \alpha \end{aligned} \quad \checkmark$$

$\therefore \angle EDB = \angle FED$ hence $\checkmark$

$\triangle FED$ is isosceles $\therefore BF = FD$. $\checkmark$

(-1) no reason each time.